

UK Visa Application Guidance & Document Readiness Agent

Demo Script

Demo Objective

The purpose of this demonstration is to show that the UK Visa Guidance Agent is more than a basic question-and-answer chatbot.

The demonstration uses five short, controlled interactions. Each interaction has been selected to show a different capability developed during the project:

1. Retrieval-augmented generation
2. Planning and tool usage
3. Conversation memory
4. Adaptive behaviour
5. Safety and graceful failure

Interaction 1 — RAG and Official Guidance

User Input

What documents are required for a Skilled Worker visa?

Expected Behaviour

The agent should recognise this as a UK visa guidance question.

It should search the ChromaDB vector store for relevant sections of the indexed GOV.UK documents.

The workflow should be:

User question → Retrieval → Relevant document chunks → Gemini → Final answer

The response should summarise the information found in the retrieved guidance rather than simply providing a generic answer from the model.

UK_Visa_Guidance_Agent.ipynb

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Commands + Code + Text Run all

[61] Os

```
question = "What documents are required for a Skilled Worker visa?"

query_vector = embed_batch([question])[0]

results = collection.query(
    query_embeddings=[query_vector],
    n_results=3
)

for i, doc in enumerate(results["documents"][0]):

    print(f"*80)
    print(f"Result {i+1}")
    print(f"*80)

    print(doc[:700])
```

...

Result 1

Validity requirements for a Skilled Worker
SW 1.1. A person applying for entry clearance or permission to stay as a Skilled Worker must apply online on the gov.uk website on the specified form as follows:
(a) for applicants outside the UK, form "Skilled Worker visa"; or
(b) for applicants inside the UK, form "Skilled Worker".

UK_Visa_Guidance_Agent.ipynb

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...

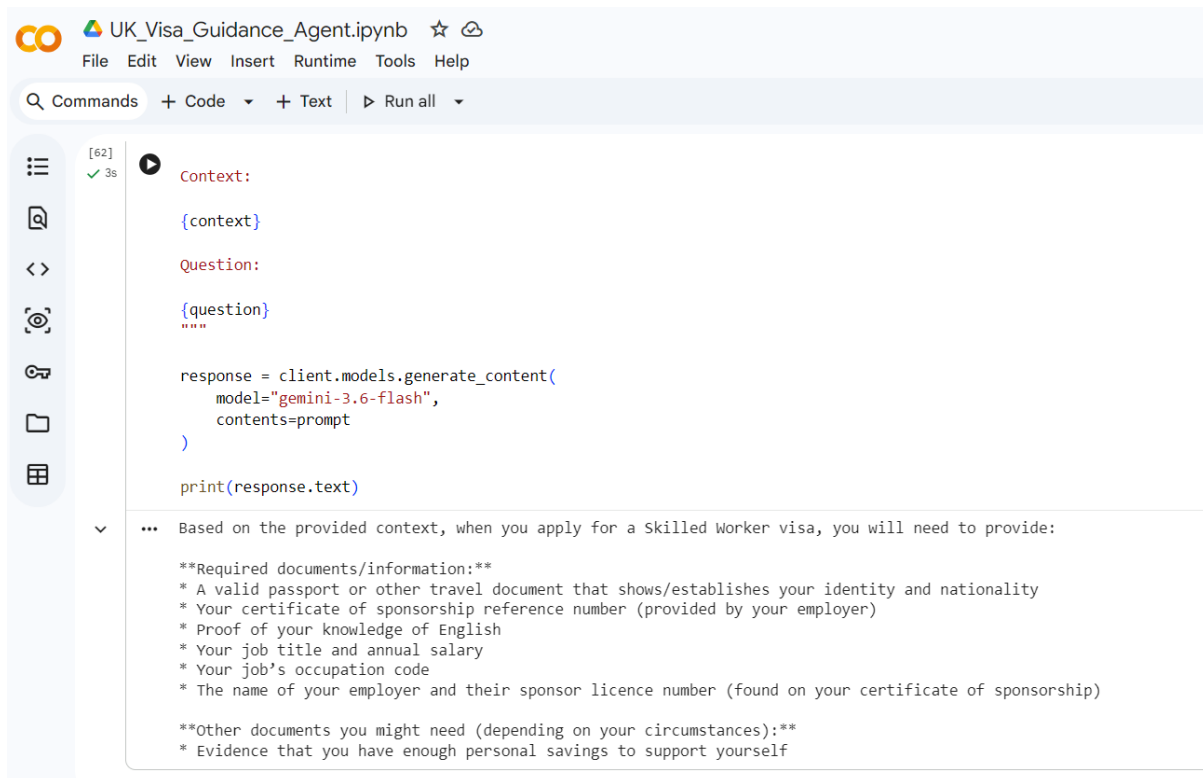
when required; and
(c) the applicant must have provided a passport or other travel document which satisfactorily establishes their

Result 2

7. Documents you'll need to apply
When you apply you'll need to provide:
your certificate of sponsorship reference number - your employer will give you this
proof of your knowledge of English (/skilled-worker-visa/knowledge-of-english)
a valid passport or other document that shows your identity and nationality
your job title and annual salary
your job's occupation code
the name of your employer and their sponsor licence number - this will be on your certificate of sponsorship
Ask your employer for a copy of your certificate of sponsorship if you do not have one.
Other documents you might need
Depending on your circumstances, you might be asked to provide:
evidence that you have enough person

Result 3

Skilled Worker visa
1. Overview
A Skilled Worker visa allows you to come to or stay in the UK to do an eligible job with an approved employer.
This visa has replaced the Tier 2 (General) work visa.
It also allows you to work in UK waters.
Eligibility



The screenshot shows a Jupyter Notebook titled "UK_Visa_Guidance_Agent.ipynb". The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar with icons for commands, code, text, and running. The notebook contains a Python script that defines a context and a question, then uses the Google Gemini API to generate a response. The output of the script is displayed below the code cell.

```
[62] ✓ 3s
Context:

{context}

Question:

{question}
"""

response = client.models.generate_content(
    model="gemini-3.6-flash",
    contents=prompt
)

print(response.text)
```

... Based on the provided context, when you apply for a Skilled Worker visa, you will need to provide:

****Required documents/information:****

- * A valid passport or other travel document that shows/establishes your identity and nationality
- * Your certificate of sponsorship reference number (provided by your employer)
- * Proof of your knowledge of English
- * Your job title and annual salary
- * Your job's occupation code
- * The name of your employer and their sponsor licence number (found on your certificate of sponsorship)

****Other documents you might need (depending on your circumstances):****

- * Evidence that you have enough personal savings to support yourself

What This Demonstrates

This interaction demonstrates:

- Document ingestion
- Text chunking
- Embeddings
- Semantic search
- ChromaDB
- RAG
- Grounded response generation

"The important part of this example is that the answer is connected to the project's official guidance knowledge base rather than relying only on the model's general knowledge."

Interaction 2 — Planning and Tool Selection

User Input

I have a sponsor, passport and English qualification. Am I eligible and what documents do I need?

Expected Behaviour

The agent should recognise that the question contains two related tasks.

It should use the basic eligibility tool and then retrieve relevant documentation from the knowledge base.

The expected flow is:

User question → Planning → Eligibility tool → Document search → Final response

The agent should combine the results into one understandable answer.

```
[87] ✓ 10s print(visa_agent("I have a sponsor, passport and English qualification. Am I eligible and what documents do I need? "))

... 2026-07-25 21:17:39,443 | INFO | Question: I have a sponsor, passport and English qualification. Am I eligible and what documents do I need?
INFO:VisaAgent:Question: I have a sponsor, passport and English qualification. Am I eligible and what documents do I need?
=====
PLAN
{'steps': ['check_basic_eligibility', 'search_documents', 'answer_directly']}
=====
2026-07-25 21:17:49,637 | INFO | Answer: Please note that any eligibility check mentioned is **only an initial assessment**. Based on the information provided, you appear to satisfy the basic requirements for a Skilled Worker visa.

---

### Eligibility Assessment
Having a valid passport, a sponsor (Certificate of Sponsorship), and proof of your English qualification covers several core requirements for a Skilled Worker visa. To be fully eligible, you must also meet the following criteria:

* **Age Requirement (SW 1.3):** You must be aged **18 or over** on the date of application.
* **Certificate of Sponsorship (CoS) Timing:** Your CoS must have been issued by your sponsor no more than **3 months (or 90 days)** before the date of application.
* **Government/International Scholarship Consent (SW 1.4):** If you received an award covering both fees and living costs from a government or international scholarship agency, you must provide written consent from that agency.

---

### Documents You Will Need to Provide
Based on the official guidance, here are the required and potential documents you must submit when applying:

#### Required Documents:
* **Valid Passport:** Or another travel document that satisfactorily establishes your identity and nationality.
* **Certificate of Sponsorship Reference Number:** Provided by your employer/sponsor.
* **Proof of Knowledge of English:** Your qualifying English language test, degree, or evidence of exemption.
* **Job Details:** Information showing your job title and annual salary.
* **Occupation Code:** Your job's specific occupation code.

#### Additional Documents You Might Need (Depending on Circumstances):
* **Financial Evidence:** Evidence that you have enough personal savings to support yourself in the UK (unless your employer certifies maintenance on your CoS).
* **Scholarship Agency Consent:** Written consent from your sponsoring agency if you received government/international scholarship funding in the past 12 months.

2026-07-25 21:17:49,639 | INFO | Latency: 10.19 seconds
INFO:VisaAgent:Latency: 10.19 seconds
Response Time: 10.19 seconds
Please note that any eligibility check mentioned is **only an initial assessment**. Based on the information provided, you appear to satisfy the basic requirements checked by the system.
```

This demonstrates:

- Planning
- Task decomposition
- Tool selection
- Tool execution
- RAG
- Multi-step agent behaviour

The eligibility tool is only a basic preliminary check. It should not be presented as a complete immigration eligibility assessment or guarantee of a successful visa application.

Interaction 3 — Conversation Memory

Start with:

User

What documents are required for a Skilled Worker visa?

Allow the agent to respond.

Then ask:

User

What about my wife?

Expected Behaviour

The second question does not repeat the visa category. The agent should use the previous conversation context to understand that the question is about the dependant of a Skilled Worker visa applicant.

The agent should therefore provide a relevant answer or, if the available context is insufficient, ask an appropriate clarification question.

```
[88]
✓ 21s
print(visa_agent(
    "What documents do I need for a Skilled Worker visa?"
))

print(visa_agent(
    "What about my wife?"
))

...
2026-07-25 21:22:21,160 | INFO | Question: What documents do I need for a Skilled Worker visa?
INFO:VisaAgent:Question: What documents do I need for a Skilled Worker visa?
=====
PLAN
{'steps': ['search_documents', 'answer_directly']}
=====
2026-07-25 21:22:30,708 | INFO | Answer: Please note that any eligibility check mentioned is **only an initial assessment**.

Based on official GOV.UK guidance, when applying for a Skilled Worker visa (which allows you to work in an eligible job with an approved employer in the UK or UK waters, i

Here are the specific documents and details you need to provide:

### Required Documents and Information
* **Valid Passport or Travel Document:** Must satisfactorily establish your identity and nationality.
* **Certificate of Sponsorship (CoS) Reference Number:** Provided by your approved UK employer (ask your employer for a copy if you do not have one).
* **Employer Details:** The name of your employer and their sponsor licence number (found on your Certificate of Sponsorship).
* **Job Title and Annual Salary:** Details showing your confirmed job offer meets the specific job and minimum salary requirements.
* **Job's Occupation Code:** The specific code for your eligible occupation.
* **Proof of Knowledge of English:** Mandatory proof meeting the English language requirement.

---

[88]
✓ 21s
print(visa_agent(
    "What documents do I need for a Skilled Worker visa?"
))

print(visa_agent(
    "What about my wife?"
))

...
Evidence of Personal Savings:** Depending on your circumstances, you may need to provide evidence showing you have enough personal savings to support yourself in the UK (unl
Application Fees and Biometrics:** You must pay the specified application fee and Immigration Health Charge, as well as provide your biometrics when required as part of the
-07-25 21:22:30,710 | INFO | Latency: 9.55 seconds
:VisaAgent:Latency: 9.55 seconds
-07-25 21:22:30,712 | INFO | Question: What about my wife?
:VisaAgent:Question: What about my wife?
onse Time: 9.55 seconds
se note that any eligibility check mentioned is **only an initial assessment**.

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Employer Details:** The name of your employer and their sponsor licence number (found on your Certificate of Sponsorship).
Job Title and Annual Salary:** Details showing your confirmed job offer meets the specific job and minimum salary requirements.
Job's Occupation Code:** The specific code for your eligible occupation.
Proof of Knowledge of English:** Mandatory proof meeting the English language requirement.
```

This interaction demonstrates:

- Short-term memory
- Multi-turn conversation
- Context retention
- Better follow-up handling

"Without memory, the user would have to repeat the visa type every time they asked a follow-up question. Memory makes the interaction more natural."

Interaction 4 — Adaptive Behaviour

First provide feedback:

User

Please answer using short bullet points.

Then ask:

User

What documents do I need for a Student visa?

Expected Behaviour

The agent should produce a response using short bullet points rather than its previous response format.

The feedback should be stored and used when generating the subsequent response.

```
941 print(visa_agent(  
168     "What documents are needed for a Student visa?"  
    ))  
  
    # User feedback  
    store_feedback(  
  
        question="What documents are needed for a Student visa?",  
  
        answer="...",  
  
        rating="negative",  
  
        comments="Please answer briefly with bullet points."  
    )  
  
    # Same question again  
    print(visa_agent(  
        "What documents are needed for a Student visa?"  
    ))
```

...

2026-07-25 22:14:08,168 | INFO | Question: What documents are needed for a Student visa?
INFO:VisaAgent:Question: What documents are needed for a Student visa?
=====

PLAN
{'steps': ['search_documents', 'answer_directly']}

=====

2026-07-25 22:14:15,868 | INFO | Answer: Based on official GOV.UK guidance, here are the required documents and evidence for a Student visa:


```
[94]
✓ 16s

...
**If you are 16 or 17 years old (under 18):**
* Written consent from both parents or legal guardians (or one parent if sole responsibility) covering:
  * Your visa application
  * Your living and care arrangements in the UK
  * Your travel to the UK
* A copy of your birth certificate (or another government-issued document) showing the names of your parents or legal guardians

***

**Sources:**
* GOV.UK: Student visa overview (`/student-visa`)
* GOV.UK: Documents you must provide (`/student-visa/documents-you-must-provide`)
Feedback stored.
=====
PLAN
{'steps': ['search_documents', 'answer_directly']}
=====
2026-07-25 22:14:24,386 | INFO | Answer: Based on official GOV.UK guidance, here are the required documents and evidence for a Student visa:

**Required for all applicants:**
* A current valid passport or travel document (scanned using the 'UK Immigration: ID Check' app or taken to a visa application centre)
* Proof of a course offer from a licensed student sponsor (Confirmation of Acceptance for Studies / CAS)
* Proof of sufficient money to support yourself and pay for your course
* Proof of English language ability (speaking, reading, writing, and understanding)

**Additional documents (depending on your circumstances):**
* Written consent from your financial sponsor if you received sponsorship for course fees and living costs in the last 12 months

UK_Visa_Guidance_Agent.ipynb
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...
* GOV.UK: Visa application centre search (`/find-a-visa-application-centre`)
2026-07-25 22:14:24,389 | INFO | Latency: 8.52 seconds
INFO:VisaAgent:Latency: 8.52 seconds
Response Time: 8.52 seconds
Based on official GOV.UK guidance, here are the required documents and evidence for a Student visa:

**Required for all applicants:**
* A current valid passport or travel document (scanned using the 'UK Immigration: ID Check' app or taken to a visa application centre)
* Proof of a course offer from a licensed student sponsor (Confirmation of Acceptance for Studies / CAS)
* Proof of sufficient money to support yourself and pay for your course
* Proof of English language ability (speaking, reading, writing, and understanding)

**Additional documents (depending on your circumstances):**
* Written consent from your financial sponsor if you received sponsorship for course fees and living costs in the last 12 months

**If you are 16 or 17 years old (under 18):**
* Written consent from both parents or legal guardians (or one parent if sole responsibility) covering:
  * Your visa application
  * Your living and care arrangements in the UK
  * Your travel to the UK
* A copy of your birth certificate (or another government-issued document) showing your parents' names

***

**Sources:**
* GOV.UK: Student visa overview (`/student-visa`)
* GOV.UK: Documents you must provide (`/student-visa/documents-you-must-provide`)
* GOV.UK: course guidance (`/student-visa/course`)
* GOV.UK: Money guidance (`/student-visa/money`)
* GOV.UK: Knowledge of English guidance (`/student-visa/knowledge-of-english`)
* GOV.UK: Points-based system student route guidance (`/government/publications/points-based-system-student-route`)
* GOV.UK: Visa application centre search (`/find-a-visa-application-centre`)
```

This demonstrates:

- Feedback collection
- Feedback storage
- Behaviour adjustment
- Before-and-after response comparison

"The agent is not changing the underlying visa information. It is changing how the information is presented based on explicit user feedback."

Interaction 5 — Safety and Graceful Failure

Test A — Missing Information

Ask:

Tell me about Mars visas.

Expected Behaviour

The agent should not make up a fictional visa category.

It should explain that it cannot find relevant information in the available UK visa guidance.

```
[95]
✓ 6s print(visa_agent("Tell me about Mars visas."))

... 2026-07-25 22:30:40,258 | INFO | Question: Tell me about Mars visas.
INFO:VisaAgent:Question: Tell me about Mars visas.
=====
PLAN
{'steps': ['answer_directly']}
=====
2026-07-25 22:30:46,520 | INFO | Answer: No relevant GOV.UK guidance found for "Mars visas".

* The UK Government and Home Office only issue visas for entry to and stay within the United Kingdom.
* There are currently no official UK visa routes, regulations, or guidance for travel to Mars.

***

**Sources:**
* GOV.UK: Visas and immigration (`/browse/visas-immigration`)
INFO:VisaAgent:Answer: No relevant GOV.UK guidance found for "Mars visas".

* The UK Government and Home Office only issue visas for entry to and stay within the United Kingdom.
* There are currently no official UK visa routes, regulations, or guidance for travel to Mars.

***

**Sources:**
* GOV.UK: Visas and immigration (`/browse/visas-immigration`)
2026-07-25 22:30:46,521 | INFO | Latency: 6.26 seconds
INFO:VisaAgent:Latency: 6.26 seconds
Response Time: 6.26 seconds
No relevant GOV.UK guidance found for "Mars visas".
```

Test B — Out-of-Domain Request

Ask:

What is the weather in London today?

Expected Behaviour

The agent should recognise that the question is outside the intended UK visa guidance scope.

It should not pretend to have weather information or use an inappropriate visa tool.

```
[96]
✓ 5s print(visa_agent("What is the weather in London today?"))

... 2026-07-25 22:31:52,230 | INFO | Question: What is the weather in London today?
INFO:VisaAgent:Question: What is the weather in London today?
=====
PLAN
{'steps': ['answer_directly']}
=====
2026-07-25 22:31:58,203 | INFO | Answer: No relevant GOV.UK guidance found for London weather.

* The UK Visa and Immigration guidance covers entry requirements, visas, and citizenship rules, not current weather or climate forecasts.
* For official UK weather updates, please consult the UK Met Office.

***

**Sources:**
* GOV.UK: Visas and immigration (`/browse/visas-immigration`)
INFO:VisaAgent:Answer: No relevant GOV.UK guidance found for London weather.

* The UK Visa and Immigration guidance covers entry requirements, visas, and citizenship rules, not current weather or climate forecasts.
* For official UK weather updates, please consult the UK Met Office.

***

**Sources:**
* GOV.UK: Visas and immigration (`/browse/visas-immigration`)
2026-07-25 22:31:58,205 | INFO | Latency: 5.97 seconds
INFO:VisaAgent:Latency: 5.97 seconds
Response Time: 5.97 seconds
No relevant GOV.UK guidance found for London weather.
```

These tests demonstrate:

- Domain restrictions
- Missing-information handling
- Hallucination prevention
- Tool-selection safeguards
- Graceful failure

Demo Summary

The first interaction shows that the agent can use a knowledge base rather than relying only on the LLM.

The second demonstrates that it can plan a multi-step request and select appropriate tools.

The third demonstrates conversation memory.

The fourth shows that the agent can use feedback to change its behaviour.

The final interaction demonstrates that the system has boundaries and does not need to answer every question simply because it can generate text.

The project is intentionally positioned as a **UK visa guidance and document-readiness assistant**, rather than a system that makes immigration decisions.

Closing Statement

"The main objective of this project was to build an agent that can help users navigate official UK visa information more efficiently. The agent combines retrieval, tool use, planning, memory and adaptive behaviour to make the interaction more useful than a basic chatbot. At the same time, the system is designed with clear limitations because UK immigration guidance can change and an AI system should not be treated as a substitute for official guidance or professional immigration advice."